

GENERAL SPECIFICATION G3.1

General

CONTENTS

G3.1 General.....1

 G3.1.1 Scope.....1

 G3.1.2 Materials1

 G3.1.3 Level Delivery, Storage and Handling1

 G3.1.4 Samples and Tests.....1

 G3.1.5 Samples of Workmanship2

 G3.1.6 Testing Facilities.....2

 G3.1.7 Setting Out3

G3.1 GENERAL

G3.1.1 Scope

- (a) This section of the Specification is applicable to the civil engineering and building construction works included in the Contract. The general provisions of G1 of the Specification shall also apply to these works.

G3.1.2 Materials

- (a) The term 'materials' shall mean all materials and articles of every kind whether raw, processed or manufactured to be supplied by the Contractor for incorporation in the Works.

G3.1.3 Level Delivery, Storage and Handling

- (a) All materials shall be delivered and stored in such a way that they are not damaged during delivery, off-loading, handling, storing or use in the Works. In general, the period of storage on Site shall be kept to the minimum possible consistent with the progress of the Works. Appropriate weatherproof and secure stores shall be provided to ensure compliance with this general requirement. A stock control system shall be implemented to ensure that materials are used in rotation as delivered.
- (b) In particular, materials which are delivered in bulk in a similar manner, for example by tipping lorry, shall be properly segregated to prevent intermixing of different grades, and shall be stored on a suitable hard surface to prevent contamination from the underlying soil. If this is not practicable for a particular operation, care shall be taken during handling subsequent to delivery and storage, to avoid contamination from the underlying soil.
- (c) Materials which are subject to deterioration from damp or humidity shall be delivered in packages which exclude moisture. Once opened, such materials shall be used on the same day as opening. Any unused material shall be disposed of at the end of the day.
- (d) Proprietary materials shall be stored in accordance with the manufacturer's instructions. Those with limited shelf life shall be used within the specified period. Any which are unused by the shelf life date shall be rejected and shall be disposed of off the Site.
- (e) Any other materials which become damaged or deteriorate in other ways shall be rejected and shall be disposed of off the Site.

G3.1.4 Samples and Tests

- (a) The Contractor shall submit samples of such materials as may be required by the S.O. and shall carry out the specified tests directed by the S.O. at the place of manufacture or on the Site or at a laboratory approved by the S.O..
- (b) Samples shall be submitted and tests carried out sufficiently early to enable further samples to be submitted and tested if required by the S.O.. The Contractor shall prepare the necessary test pieces and supply all labour, appliances, testing apparatus and everything necessary for the carrying out of all the specified tests.

- (c) Unless otherwise provided for separately in the Bill of Quantities the cost of the samples and of conducting all the specified tests on the samples shall be included in the Contract Rates.
- (d) The Contractor shall give the S.O. fourteen (14) days' notice in writing of the date on which any of the samples will be ready for testing or inspection at the place of manufacture or at a laboratory approved by the S.O. and unless the S.O. or his representative shall attend at the appointed place within the said fourteen (14) days the test may proceed in his absence. Provided that the Contractor shall in any case submit to the S.O. within seven (7) days of every test such number of certified copies (not exceeding six) of the test readings as the S.O. may require.
- (e) Approval by the S.O. as to the placing of orders for materials or as to samples or tests shall not prejudice any of the S.O.'s power under the Contract.

G3.1.5 Samples of Workmanship

- (a) Where samples of workmanship are required - masonry and brickwork for example - the sample panels shall be located at suitable places on the Site as approved by the S.O.. Such samples of workmanship shall be offered for inspection and approval by the S.O.. Any which are unsatisfactory shall be removed and the sample panel construction repeated. The cost of the materials and workmanship in the samples, including any which are required to be repeated because of rejection, shall be deemed to be included in the Contract Rates.
- (b) After approval by the S.O. these samples shall be protected and preserved until the work they represent has been completed, or otherwise as agreed with the S.O.. Workmanship in progress shall be compared with the approved samples, and any which is found unsatisfactory shall be rejected.

G3.1.6 Testing Facilities

- (a) The Contractor shall at his own expense arrange with the suppliers or manufacturers to supply the necessary gauges and prepare all test pieces and supply all labour and apparatus for testing which may be necessary or which may be required by the S.O. or the Inspectors for carrying out the tests and requirements of this Specification at the Contractor's premises or at the place of manufacture, and shall also provide all test pieces required. Should the Contractor fail to perform any of the foregoing obligations, the S.O. shall be at liberty to perform the said obligations or any of them which the Contractor has failed to perform, either at the Contractor's premises, places of manufacture or elsewhere and charge the costs and expenses thereof to the Contractor.
- (b) The Contractor may propose a third party specialist laboratory for the purpose of testing concrete and soils for the approval of the S.O.. Alternatively, the Contractor shall establish on the Site within two (2) weeks of the start of the Contract a laboratory for the purposes of testing concrete and soils used on the Site complete with a designated technician as specified.
- (c) The Site Laboratory shall be housed in a single room designated solely for this purpose, having a floor area not less than 15 square metres and which is provided with all work benches, cupboards, furnishing materials and equipment necessary for the purposes of carrying out the tests specified in the general specifications.

G3.1.7 Setting Out

- (a) The Contractor shall be responsible for accurately setting out the works to the specified positions, dimensions, levels, and Building Lines and also checking the site survey for dimensional and level accuracy and reporting any discrepancies before construction and building work commences. Any errors in position, level, dimension or alignment of any part of the works at any time shall be rectified by the Contractor at his own expense. The Contractor shall provide the S.O. with all facilities, equipment and labour to enable him to check the setting out and levels of the works at all times. The checking of any setting out point, line or level by the S.O. shall not in any way relieve the Contractor of his responsibility. All setting out points, benchmarks, site rails, pegs and other survey points shall be clearly marked and protected from damage or disturbance during the execution of the works.
- (b) All setting out of the works shall be carried out relative to the Survey 21 system.
- (c) The Contractor shall use GPS to set out the works and shall not use historic benchmarks.
- (d) On contract award the Contractor shall establish control stations. Prior to commencing construction of the works the Contractor shall satisfy himself that all existing ground levels and all levels and locations of structures where they are relevant to the works as indicated on the drawings are correct. If the Contractor considers that there is any inconsistency within the information given, the matter shall be referred to the S.O. for resolution before work commences.
- (e) The Contractor shall be responsible for setting out the works and for the correctness of the position and dimensions of all parts of the works.
- (f) Before work commences on Site the Contractor shall submit proposed methods of setting out and dimensional measurements for the construction to the S.O. for acceptance.
- (g) All survey reference stations shall be carefully preserved except where construction requires their removal. Before such removal substitute stations shall be established by the Contractor to the satisfaction of the S.O..
- (h) The Contractor shall supply to the S.O., a minimum of two weeks prior to commencement of each element of the work, records in accepted form relating to all reference pegs and works bench marks and shall keep records up to date by formal notice to the S.O..

END OF SECTION